

Analysis of cross-orientation responses as a function of depth in primary visual cortex in macaque, a two-photon imaging study

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Abstract

We used two-photon calcium imaging with GCaMP6s to measure cross-orientation interactions as a function of cortical depth in layer 2/3 of macaque V1, recording from 4,785 neurons across 28 fields of view spanning 140–518 μm below the pial surface. For each neuron, we computed the ratio P of the observed plaid response to a linear prediction from the grating tuning curve and the shape correlation C between predicted and observed plaid tuning curves. P decreased with depth ($r = -0.706$, $p = 2.7 \times 10^{-5}$, $n = 28$ sites), transitioning from facilitation ($P > 1$) in superficial layer 2/3 to suppression ($P < 1$) at depth, while C increased ($r = +0.799$, $p = 3.4 \times 10^{-7}$), indicating that the plaid tuning shape becomes more linearly predictable from the grating response even as the overall amplitude decreases. The P -depth correlation survived partial-correlation controls for ROI morphology, signal-to-noise ratio, orientation selectivity, and spatial frequency preference ($r = -0.448$, $p = 0.017$). Spatial frequency preference and local orientation homogeneity each mediated a substantial fraction of the depth gradient, consistent with broader normalization pools at greater cortical depth. The tissue is being prepared for serial electron microscopy reconstruction to relate the observed depth gradient in normalization to inhibitory circuit architecture.

1 Introduction

The response of neurons in primary visual cortex (V1) to an optimally oriented grating can be suppressed by superimposing a second grating at the orthogonal orientation, a phenomenon termed cross-orientation suppression (Morrone et al., 1982; Bonds, 1989; Carandini et al., 1997; Freeman et al., 2002). Cross-orientation suppression has been studied extensively in cat and macaque V1 since the original descriptions of orientation selectivity (Hubel and Wiesel, 1962, 1968) and has been central to understanding how cortical circuits combine and normalize multiple inputs.

Several mechanisms have been proposed. Intracortical models attribute suppression to orientation-tuned inhibitory interneurons that directly suppress responses to non-preferred orientations (Sillito, 1975; Bonds, 1989; Shapley et al., 2003). Feedforward models attribute it to broadly tuned inhibition arriving from the lateral geniculate nucleus (Priebe and Ferster, 2006; Li et al., 2006). Divisive normalization models explain suppression as a consequence of the neuron’s response being divided by a weighted sum of activity across a normalization pool: when both component gratings drive the pool, the denominator is large, and the response to the preferred grating is reduced (Heeger, 1992; Carandini and Heeger, 2012). These mechanisms are not mutually exclusive, and the normalization framework in particular accounts quantitatively for cross-orientation suppression in both simple and complex cells (Carandini et al., 1997), explains contrast gain control (Albrecht and Hamilton, 1982; Sclar et al., 1990; Schwartz and Simoncelli, 2001), and generalizes to attentional modulation (Reynolds and Heeger, 2009).

Nearly all prior work on cross-orientation suppression has relied on single-unit electrophysiology, recording from one neuron at a time (DeAngelis et al., 1992; Carandini et al., 1997; Smith et al., 2006; Busse et al., 2009). Although this approach has yielded precise characterizations of suppression in individual cells, it leaves open how suppression and facilitation are distributed across neural populations and, in particular, whether they vary systematically with cortical depth. The majority of experiments have been conducted in cat V1, with comparatively fewer studies in macaque (Carandini et al., 1997; Smith et al., 2006), the species whose cortical organization most closely resembles that of humans.

Layer 2/3 of macaque V1 spans several hundred micrometers and contains sublayers with distinct connectivity patterns, cell-type distributions, and functional properties (Lund, 1988; Callaway,

1998; Douglas and Martin, 2004; Briggs, 2010). Superficial portions receive strong input from layer 4C and participate in local recurrent circuits, while deeper portions connect more extensively with layer 5 and other cortical layers (Callaway, 1998; Douglas and Martin, 2004). The functional architecture is organized into orientation columns and pinwheel structures (Blasdel and Salama, 1986; Bonhoeffer and Grinvald, 1991; Ohki et al., 2005, 2006), and local homogeneity of orientation preference, spatial frequency tuning, and orientation selectivity all contribute to the local circuit context in which cross-orientation interactions occur (Nauhaus et al., 2008; Ringach et al., 2002). Whether cross-orientation interactions change systematically across the depth range of layer 2/3 has not been tested at the population level.

Two-photon calcium imaging with genetically encoded indicators (Denk et al., 1990; Stosiek et al., 2003; Chen et al., 2013; Grienberger and Konnerth, 2012) provides the combination of cellular resolution and population-scale recording needed to address this question, and recent viral delivery strategies for nonhuman primates have made such recordings feasible (Deverman et al., 2016; Li et al., 2017; Tang et al., 2018).

Here, we used two-photon calcium imaging with GCaMP6s to measure cross-orientation interactions across approximately 400 μm of cortical depth in layer 2/3 of macaque V1. We recorded from 4,785 neurons across 28 fields of view spanning 140–518 μm below the pial surface and, for each neuron, compared the observed response to orthogonal plaid stimuli with a linear prediction derived from the grating tuning curve, quantifying suppression and facilitation with two complementary metrics. We found that cross-orientation interactions transition from facilitation in superficial layer 2/3 to suppression at greater depth, consistent with stronger normalization in deeper sublayers, and that this depth gradient is partially mediated by spatial frequency preference and orientation tuning bandwidth.

The long-term goal of this project is to combine large-scale functional imaging with dense connectomics from serial electron microscopy of the same tissue to relate circuit wiring to normalization in primate V1 (Chatterjee et al., 2021).

2 Methods

2.1 Neurophysiology

Two-photon calcium imaging was performed in primary visual cortex (V1) of one anesthetized pigtail macaque (*Macaca nemestrina*). The calcium indicator GCaMP6s was delivered via injection of the viral vector PHP.eB-CAG-GCaMP6s into V1. A cranial imaging window was implanted over the injection site, and the animal was allowed a 26-day recovery period for viral expression. Terminal recording sessions were conducted over 5 consecutive days under general anesthesia with standard physiological monitoring.

Two-photon imaging was performed using a modified Sutter MOM (Movable Objective Microscope) system equipped with a Spectra-Physics MaiTai Ti:sapphire laser. Image acquisition was controlled by ScanImage software. Single-plane scanning was performed at 30 Hz frame rate. Laser power was adjusted for each imaging depth to compensate for increased scattering at greater depths while staying below the threshold for photodamage.

A total of 28 fields of view (FOV) were acquired at depths ranging from 140 to 518 μm below the pial surface, spanning the extent of layer 2/3. Each FOV covered a 512×512 pixel area. Regions of interest (ROIs) corresponding to individual neurons were identified using Suite2p (Pachitariu et al., 2017), yielding approximately 4,785 ROIs across all sites (mean $\pm$ SD: 171 ± 28 ROIs per site; range: 120–230).

2.2 Visual stimuli

Visual stimuli consisted of drifting sinusoidal gratings and orthogonal plaids. Gratings were presented at 50% contrast, 4 Hz temporal frequency, 4 cyc/deg spatial frequency, within a 2-degree circular patch centered on the aggregate receptive field of each recording site. Plaid stimuli were constructed by superimposing two gratings at orthogonal orientations, each at the same contrast, temporal frequency, and spatial frequency as the component gratings.

Stimulus direction was varied in 30-degree increments (12 directions spanning 0–330°). Each stimulus was presented for 2 s, followed by a 1.5-s mean-gray inter-stimulus interval. Each direction condition was repeated for 6 trials, and trial order was randomized within each block.

In addition to the grating and plaid stimuli, receptive field maps were obtained using flashed

light and dark spots, and spatial frequency tuning was measured at 8 spatial frequencies (0.64–7.68 cyc/deg).

2.3 Data analysis

Image registration and ROI extraction were performed using Suite2p (Pachitariu et al., 2017). For each ROI, the stimulus-evoked response was computed as the mean fluorescence in the window 200–2,000 ms after stimulus onset. The baseline fluorescence was computed from the window 250–550 ms before stimulus onset (during the preceding inter-stimulus interval). All analyses were performed on baseline-subtracted responses (ΔF).

2.3.1 Metric P: suppression/facilitation ratio

For each ROI, a linear prediction of the plaid response was constructed as follows. The baseline-subtracted grating tuning curve $G(\theta)$ was shifted by -90° and summed with the original to produce the predicted plaid tuning curve:

$$\text{Pred}(\theta) = [G(\theta) - B] + [G(\theta - 90^\circ) - B], \quad (1)$$

where B is the baseline fluorescence. The primary metric P quantifies the ratio of the observed plaid response to this linear prediction, summed across all directions:

$$P = \frac{\sum_i R_i^{\text{plaid}}}{\sum_i \text{Pred}_i}, \quad (2)$$

where R_i^{plaid} is the baseline-subtracted plaid response at direction i . Values of $P > 1$ indicate facilitation (the observed plaid response exceeds the linear prediction), $P < 1$ indicates suppression (the observed response falls below the prediction), and $P = 1$ indicates perfect linearity. Because P is a ratio, it is plotted on a logarithmic axis so that equal distances represent equal multiplicative changes and the neutral point ($P = 1$) falls at the center of the scale.

2.3.2 Metric C: shape similarity

Metric C is the Pearson correlation between the predicted and observed plaid tuning curves:

$$C = r(\text{Pred}(\theta), R^{\text{plaid}}(\theta)). \quad (3)$$

Higher values of C indicate that the shape of the plaid tuning curve is more linearly predictable from the component grating responses, independent of the overall amplitude ratio captured by P .

2.3.3 Signal-to-noise ratio

Signal-to-noise ratio (SNR) was computed separately for grating and plaid responses as:

$$\text{SNR} = \frac{\text{Var}_{\text{signal}}}{\text{Var}_{\text{noise}}}, \quad (4)$$

where signal variance was estimated from the tuning curve means across directions and noise variance from the trial-to-trial variability.

2.3.4 Additional metrics

Orientation Selectivity Index (OSI). OSI was computed using standard circular variance methods (Ringach et al., 2002) to quantify the degree of orientation tuning for each neuron.

Best spatial frequency. The spatial frequency eliciting the maximum response was determined from tuning curves measured at 8 spatial frequencies (0.64–7.68 cyc/deg) (De Valois et al., 1982).

Orientation tuning half-width. Half-width at half-height of the grating tuning curve, measured on the flank toward the orthogonal orientation. This captures how broadly the neuron responds toward the orientation of the cross-oriented grating component.

Local Homogeneity Index (LHI). LHI quantifies the similarity of orientation preference between neighboring ROIs within each field of view, computed using a Gaussian weighting kernel with $\sigma = 35 \mu\text{m}$ (Nauhaus et al., 2008). Higher LHI indicates more homogeneous local orientation preference, as found near orientation domain centers; lower LHI indicates heterogeneous preferences,

as found near pinwheel centers.

2.4 Statistical analysis

2.4.1 Site-level analysis

For primary analyses, metrics were averaged across all ROIs within each field of view to obtain site-level means ($n = 28$). This approach accounts for the hierarchical structure of the data (ROIs nested within sites) and avoids pseudoreplication.

2.4.2 Correlation and regression

Because P is a ratio bounded below by zero, all site-level correlations and partial correlations involving P were computed on $\log_2(P)$ to satisfy normality assumptions and ensure that equal multiplicative changes in P receive equal weight. Pearson correlation coefficients were computed between cortical depth and metric values. Statistical significance was assessed using two-tailed tests with $\alpha = 0.05$. Linear regression was used to estimate depth slopes. Non-parametric bootstrap (10,000 iterations, resampling sites with replacement) was used to compute 95% confidence intervals for correlation coefficients.

2.4.3 Partial correlations

To control for potential confounds, partial correlations were computed between depth and metrics while statistically controlling for ROI radius, SNR, OSI, and best spatial frequency. Partial correlations were obtained by regressing out confound variables from both depth and metric values, then correlating the residuals.

2.4.4 Mediation analysis

To identify which covariates mediate the P -depth relationship, partial correlations were computed controlling for each covariate individually. The percent reduction in the absolute correlation relative to the zero-order $r = -0.706$ quantifies the mediating influence of each variable. Because spatial frequency preference, tuning bandwidth, and local homogeneity are themselves correlated with one another and with depth, individual mediation percentages may overlap.

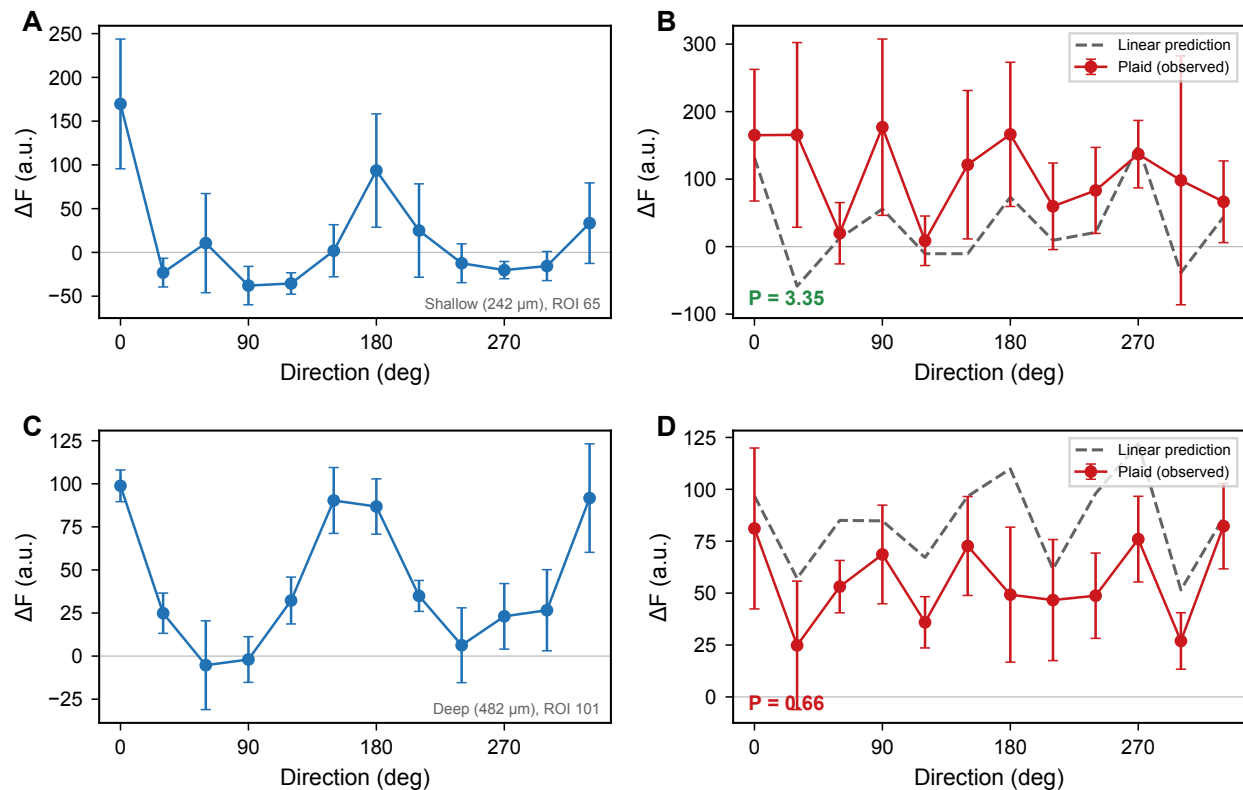


Figure 1: **Example cross-orientation analysis at shallow and deep cortical depths.** (A) Grating tuning curve (ΔF) for an example ROI at a shallow site ($242\ \mu\text{m}$). (B) Observed plaid response (red) versus linear prediction (dashed gray) for the same ROI. $P = 3.35$, indicating strong facilitation. (C) Grating tuning curve for an example ROI at a deep site ($482\ \mu\text{m}$). (D) Observed plaid response versus prediction for the deep ROI. $P = 0.66$, indicating suppression. Error bars show SEM across trials.

3 Results

3.1 Dataset overview

We recorded calcium signals from 4,785 ROIs across 28 fields of view in layer 2/3 of macaque V1, spanning cortical depths of $140\text{--}518\ \mu\text{m}$ below the pial surface (Figure 1). Each ROI was presented with drifting sinusoidal gratings and orthogonal plaids, and cross-orientation interaction metrics were computed by comparing the observed plaid response to a linear prediction derived from the grating tuning curve.

Figure 2: **Linear prediction model for plaid responses.** (A) Gabor spatial filter at the preferred orientation. (B) Temporal filter modeled as a difference of Maxwell functions (time constants $s_1 = 20$ ms, $s_2 = 40$ ms), producing a space-time separable linear filter. (C) The linear prediction is the sum of the responses to each component grating of the plaid, computed by shifting and summing the grating tuning curve. Deviations from this prediction are quantified by metrics P (ratio) and C (shape correlation).

3.2 Cross-orientation suppression increases with cortical depth

Metric P decreased with cortical depth at the site level ($r = -0.706$, $p = 2.72 \times 10^{-5}$, $n = 28$; Figure 3A), with superficial sites (140–266 μm) exhibiting $P > 1$ (facilitation: the observed plaid response exceeded the linear prediction) and deeper sites (392–518 μm) exhibiting $P < 1$ (suppression). The transition from net facilitation to net suppression occurred at approximately 350 μm depth.

3.3 Plaid tuning shape becomes more predictable with depth

Metric C , the Pearson correlation between predicted and observed plaid tuning curves, increased with depth ($r = +0.799$, 95% CI: $[+0.679, +0.892]$, $p = 3.38 \times 10^{-7}$; Figure 3B), indicating that deeper neurons have plaid tuning curve shapes that are more linearly predictable from the grating response despite showing stronger amplitude suppression (lower P). The metrics P and C were anticorrelated across sites ($r = -0.571$, $p = 0.002$; Figure 3C): as the gain of the plaid response decreased with depth, its shape became more predictable, consistent with a divisive normalization mechanism that scales gain while preserving the form of the tuning curve.

3.4 Spatial frequency preference varies with depth

Mean preferred spatial frequency decreased with cortical depth ($r = -0.785$, $p < 10^{-6}$; Figure 4), consistent with the known laminar gradient in spatial frequency preference in macaque V1 (De Valois et al., 1982). Within individual sites, spatial frequency showed weak correlations with P (significant at 7 of 28 sites), indicating that spatial frequency preference does not strongly predict P within a single imaging plane; however, across the full depth range, controlling for spatial frequency reduced the P –depth correlation from $r = -0.706$ to $r = -0.220$ ($p = 0.26$), accounting for approximately 69% of the effect (Table 1). This is consistent with a mechanism in which deeper neu-

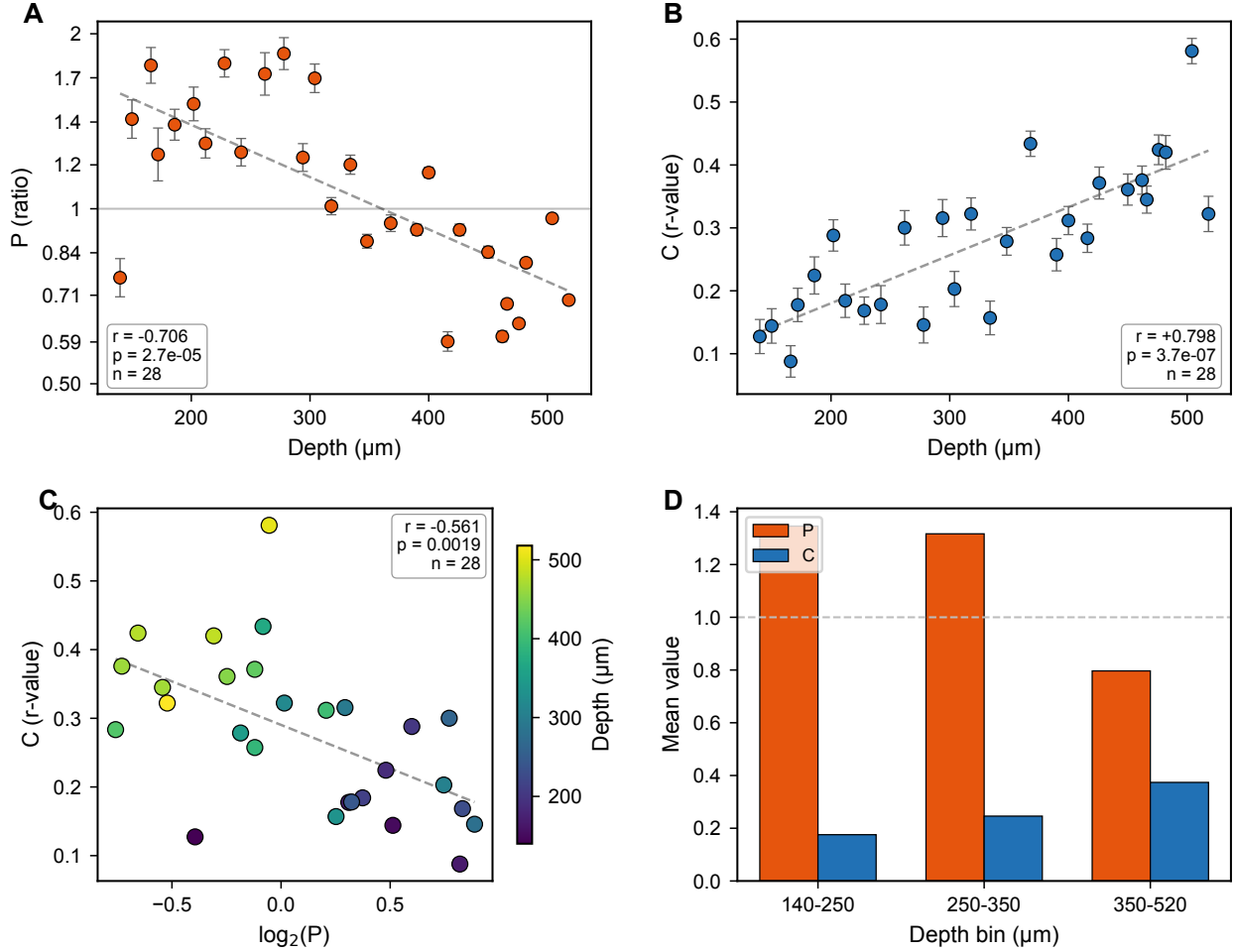


Figure 3: Depth-dependent changes in cross-orientation metrics. (A) Metric P (suppression/facilitation ratio) decreased with cortical depth ($r = -0.706$, $p = 2.72 \times 10^{-5}$, $n = 28$). Each point represents the site-mean P across all ROIs at one imaging depth, plotted on a logarithmic axis. Dashed line shows linear regression. $P > 1$: facilitation; $P < 1$: suppression. (B) Metric C (shape similarity) increased with depth ($r = +0.799$, $p = 3.38 \times 10^{-7}$). Higher C indicates the plaid tuning curve shape is more linearly predictable from grating responses. (C) Scatter plot of P versus C across sites, colored by cortical depth, showing their anticorrelation. (D) Depth-binned summary. Superficial layer 2/3 shows facilitation ($P > 1$); deep layer 2/3 shows suppression ($P < 1$).

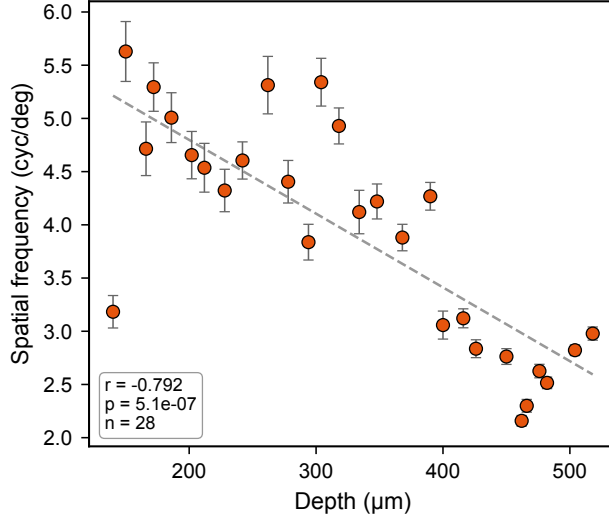


Figure 4: **Spatial frequency preference decreases with cortical depth.** Site-mean preferred spatial frequency plotted against depth. Each point is one field of view; error bars show SEM. Dashed line is the linear regression fit.

rons, preferring lower spatial frequencies, have larger receptive fields and correspondingly broader normalization pools, producing stronger cross-orientation suppression.

3.5 Baseline fluorescence increases with depth

Baseline fluorescence increased with cortical depth, consistent with the progressive increase in laser power applied at deeper imaging planes to compensate for optical scattering (Figure 5). Within individual sites, P and baseline fluorescence were uncorrelated at 24 of 28 sites, indicating that the depth gradient in P is not an artifact of imaging depth. All metrics used baseline-subtracted responses.

3.6 Orientation tuning bandwidth broadens with depth

Orientation tuning half-width increased with cortical depth ($r = +0.869$, $p = 2.04 \times 10^{-9}$; Figure 6A), with mean half-width broadening from approximately 60° at superficial sites to approximately 80° at deeper sites. Within individual sites, half-width showed a significant negative correlation with C at 15 of 28 sites (Figure 6B): at a given depth, more broadly tuned neurons tended to have less linearly predictable plaid tuning shapes. Tuning bandwidth mediated approximately 29% of the P -depth effect (Table 1).

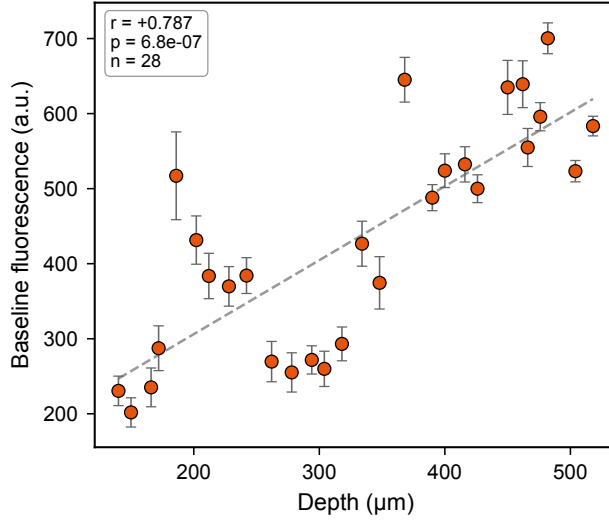


Figure 5: **Baseline fluorescence increases with depth.** Site-mean baseline fluorescence as a function of cortical depth, reflecting the increase in laser power at deeper imaging planes.

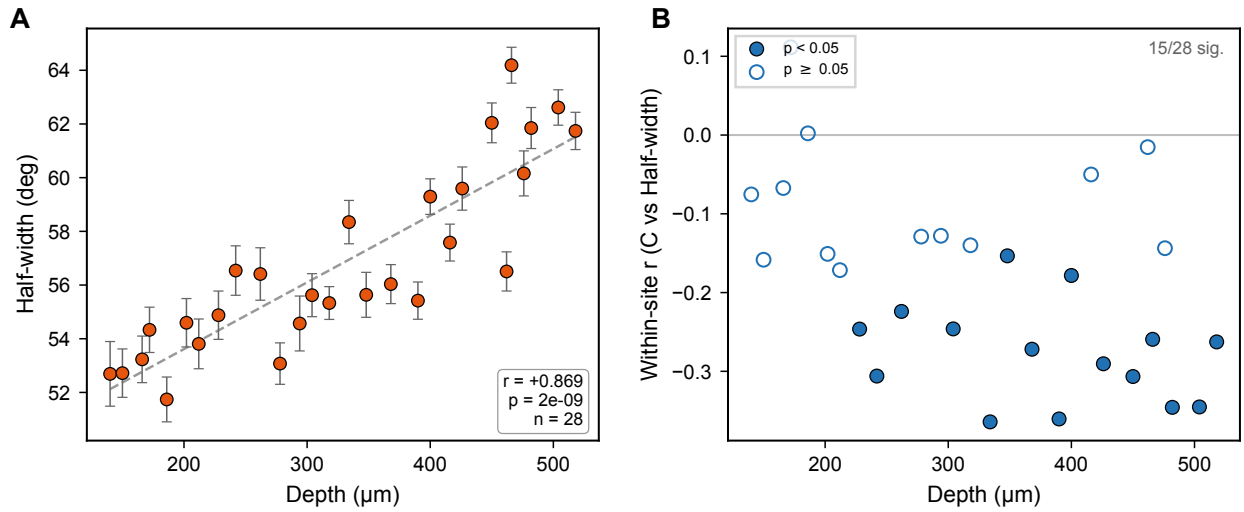


Figure 6: **Orientation tuning bandwidth increases with depth.** (A) Site-mean half-width at half-height versus depth ($r = +0.869$, $p = 2.0 \times 10^{-9}$). (B) Within-site Pearson r between half-width and C , plotted against depth. Filled circles: $p < 0.05$; open circles: $p \geq 0.05$. The consistent negative correlation shows that, within sites, more broadly tuned neurons have less predictable plaid tuning shapes.

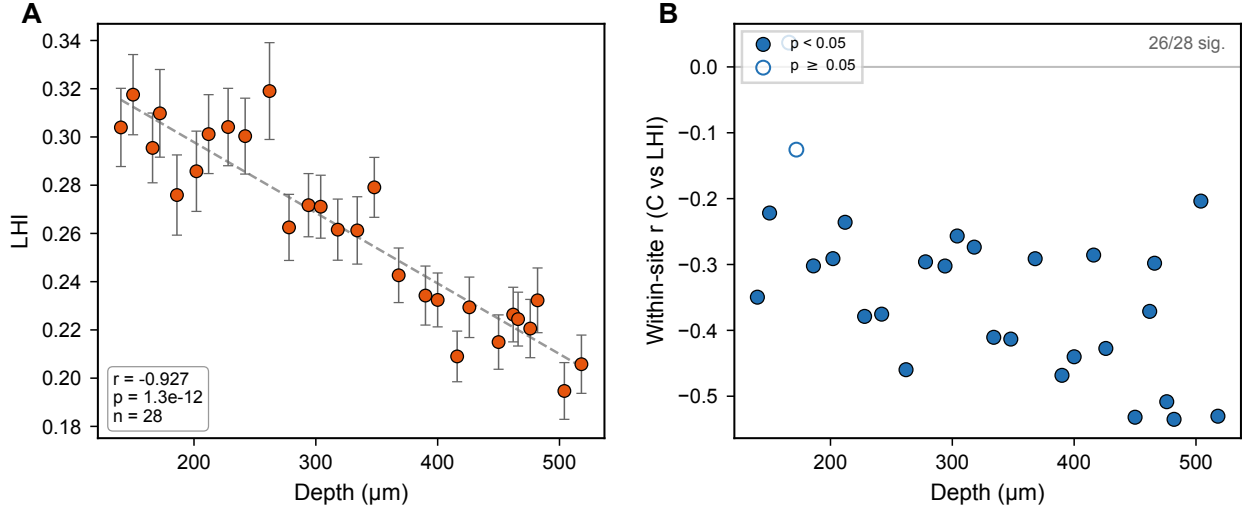


Figure 7: **Local homogeneity index decreases with depth.** (A) Site-mean LHI versus depth ($r = -0.93$). Higher LHI indicates more homogeneous local orientation preferences (orientation domains); lower LHI indicates greater heterogeneity (near pinwheels). (B) Within-site Pearson r between C and LHI, plotted against depth. A consistent negative relationship is observed at 26/28 sites.

3.7 Local homogeneity decreases with depth

The local homogeneity index (LHI) showed a strong negative correlation with depth ($r = -0.93$, $p \ll 0.001$; Figure 7A), with LHI decreasing by approximately 35% from 150 to 520 μm . This is consistent with a transition from more homogeneous orientation preference maps in superficial layer 2/3 (orientation domains) to more heterogeneous maps at depth (near pinwheel centers or in regions with less columnar organization).

Within individual sites, C and LHI showed a consistently moderate negative correlation at 26 of 28 sites (Figure 7B), indicating that neurons in more heterogeneous orientation neighborhoods (lower LHI, near pinwheel centers) had more linearly predictable plaid tuning shapes.

3.8 Orientation selectivity decreases with depth

OSI decreased significantly with cortical depth ($r = -0.490$, $p = 0.008$; Figure 8A), from approximately 0.70 at 150 μm to 0.57 at 520 μm . Within individual sites, C and OSI were strongly positively correlated at 26 of 28 sites (Figure 8B): neurons with higher orientation selectivity showed more linearly predictable plaid tuning. In contrast, P showed a significant within-site relationship with OSI at only 7 of 28 sites (Figure 8C), suggesting that the strength of cross-orientation sup-

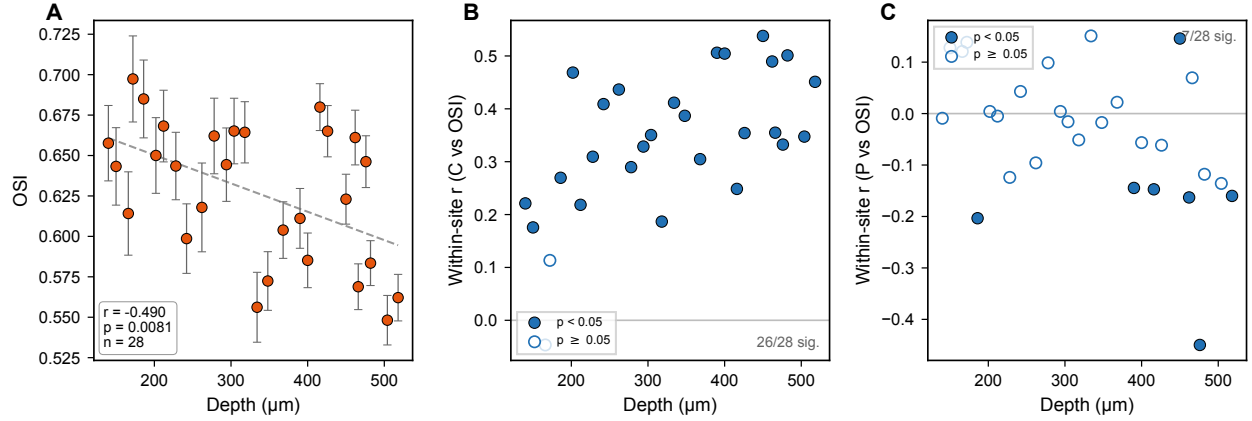


Figure 8: **Orientation selectivity and cross-orientation metrics.** (A) Site-mean OSI versus depth ($r = -0.490$, $p = 0.0081$). (B) Within-site Pearson r between C and OSI versus depth. A strong positive relationship is present at 26/28 sites: more orientation-selective neurons have more predictable plaid tuning shapes. (C) Within-site Pearson r between P and OSI versus depth. Significant at only 7/28 sites.

pression is largely independent of orientation selectivity within a given imaging plane. Consistent with this, OSI mediated less than 1% of the P -depth effect in the mediation analysis (Table 1).

3.9 Robustness and control analyses

The P -depth and C -depth correlations remained significant after controlling for potential confounds using partial correlations (Table 2). Controlling for ROI radius alone slightly strengthened the P -depth correlation (partial $r = -0.757$, $p = 3.1 \times 10^{-6}$), indicating that the depth gradient in ROI morphology does not account for the depth gradient in suppression. Simultaneous control for ROI radius, SNR, OSI, and spatial frequency preference reduced the P -depth correlation to $r = -0.448$ ($p = 0.017$), confirming that a significant depth-dependent component of the P gradient persists beyond all measured confounds. Bootstrap confidence intervals for the P -depth correlation ($r = -0.706$, 95% CI: $[-0.879, -0.440]$) excluded zero. The C -depth correlation similarly survived all controls (partial $r = +0.444$, $p = 0.018$).

3.10 Summary of depth-dependent changes

Eight metrics showed coordinated depth-dependent changes across layer 2/3 (Table 3). Superficial layer 2/3 was characterized by high spatial frequency preference, narrow orientation tuning, high selectivity, high local homogeneity, and cross-orientation facilitation. Deeper layer 2/3 showed low

Table 1: Mediation of the P -depth relationship by individual covariates. Each row shows the partial correlation between P (log-transformed) and depth after controlling for one covariate. Percent reduction is relative to the zero-order $r = -0.706$.

Control variable	Partial r	% Reduction
None (zero-order)	-0.706	—
+ LHI	-0.173	76%
+ Spatial frequency	-0.220	69%
+ Half-width	-0.500	29%
+ SNR	-0.682	3%
+ OSI	-0.705	<1%

Table 2: Partial correlations between depth and cross-orientation metrics after controlling for potential confounds ($n = 28$ sites).

Metric	Controls	Partial r	p -value
P	ROI radius	-0.757	3.1×10^{-6}
P	Radius + SNR	-0.765	2.1×10^{-6}
P	All (radius + SNR + OSI + SF)	-0.448	0.017
C	ROI radius	+0.511	0.005
C	Radius + SNR	+0.567	0.002
C	All (radius + SNR + OSI + SF)	+0.444	0.018

246 spatial frequency preference, broad tuning, lower selectivity, lower local homogeneity, and cross-
247 orientation suppression.

Table 3: Depth profile summary across layer 2/3.

Metric	Change with increasing depth
ROI size (radius, npix)	Increases
Baseline fluorescence	Increases
SNR	Increases
Orientation tuning bandwidth	Broader
Preferred spatial frequency	Lower
Orientation selectivity (OSI)	Lower
Local homogeneity (LHI)	Lower
Cross-orientation ratio (P)	More suppressive ($P < 1$)
Shape similarity (C)	More predictable

4 Discussion

We found that cross-orientation interactions transition systematically from facilitation in superficial layer 2/3 to suppression at greater cortical depth in macaque V1, with the suppression/facilitation ratio P decreasing significantly with depth ($r = -0.706$, $p = 2.72 \times 10^{-5}$, $n = 28$) and the shape-similarity metric C increasing ($r = +0.799$, $p = 3.38 \times 10^{-7}$). The P -depth effect was robust to partial-correlation controls for ROI morphology, signal-to-noise ratio, orientation selectivity, and spatial frequency preference ($r = -0.448$, $p = 0.017$), and spatial frequency and local orientation homogeneity each mediated a substantial fraction of the depth gradient.

4.1 Relation to normalization models

The observation that P decreases with depth while C increases is consistent with the divisive normalization framework (Heeger, 1992; Carandini and Heeger, 2012). If deeper neurons draw from broader normalization pools, the denominator of the normalization equation is larger, producing stronger divisive suppression (lower P) while preserving the relative shape of the tuning curve (higher C). A purely noise-driven account of the P -depth gradient would predict that both P and C degrade together; the observation that they move in opposite directions argues against this possibility and is instead consistent with a gain-scaling mechanism that reduces response amplitude while preserving tuning curve form (Carandini et al., 1997).

4.2 Spatial frequency and tuning bandwidth as mediators

Controlling for spatial frequency preference reduced the P -depth correlation by 69%, consistent with the interpretation that deeper neurons, preferring lower spatial frequencies, have larger receptive fields and correspondingly broader normalization pools (De Valois et al., 1982; Sceniak et al., 1999), which would naturally produce stronger cross-orientation suppression. Orientation tuning bandwidth mediated 29% of the effect through a related mechanism: more broadly tuned neurons at depth are less selective in which orientations contribute to the normalization pool, again strengthening suppression. Local orientation homogeneity (LHI) mediated 76% of the effect, reflecting the transition from homogeneous orientation domains in superficial layer 2/3 to more heterogeneous local structure at depth (Nauhaus et al., 2008). Because spatial frequency, bandwidth, and LHI

are themselves intercorrelated and all covary with depth, their individual mediation percentages overlap substantially. Nevertheless, the combined partial correlation controlling for radius, SNR, OSI, and SF ($r = -0.448$, $p = 0.017$) confirms that a significant component of the P -depth gradient cannot be accounted for by any of these measured covariates, suggesting that circuit-level factors—possibly the density or specificity of inhibitory connections—contribute independently to the depth gradient in normalization strength.

4.3 Comparison with prior work

Nearly all prior characterizations of cross-orientation suppression have relied on single-unit electrophysiology, recording from one neuron at a time in cat (Morrone et al., 1982; Bonds, 1989; DeAngelis et al., 1992) or macaque (Carandini et al., 1997; Smith et al., 2006) V1. These studies established that suppression is widespread but showed considerable cell-to-cell variability. Our population-level approach reveals that much of this variability is structured: it follows a systematic depth gradient across layer 2/3. This gradient was not accessible to single-electrode studies, which typically sample isolated neurons without precise depth registration.

Two-photon imaging of cross-orientation suppression has been reported previously in mouse V1 (Busse et al., 2009), but mouse V1 lacks the columnar organization of primate V1, making direct comparison difficult. The depth gradient in local orientation homogeneity that we observe—from high-LHI orientation domains superficially to low-LHI heterogeneous regions at depth—has no clear analogue in the salt-and-pepper organization of mouse V1.

4.4 The P - C dissociation

The anticorrelation between P and C across sites ($r = -0.571$, $p = 0.002$) provides an important internal control. If the depth gradient in P were driven by noise, data quality, or neuropil contamination, one would expect C to decrease in parallel, because noisier data would make tuning shapes less predictable. Instead, C increases with depth: deeper neurons have plaid tuning shapes that are more faithfully predicted by the linear model even as their overall amplitude is reduced. This pattern is difficult to explain by any artifact of imaging depth and is instead consistent with a gain-scaling mechanism—divisive normalization—that reduces response amplitude while preserving the shape of the tuning curve.

4.5 Limitations

Several limitations apply. All recordings were obtained from a single animal (*Macaca nemestrina*); however, the internal consistency of the P -depth gradient across 28 independent fields of view, acquired over 5 recording sessions, provides confidence that the effect is not driven by idiosyncratic features of a small number of sites. Generalization to other individuals and species remains to be determined. The design is correlational: we cannot establish whether the depth gradient in normalization is caused by the depth gradient in spatial frequency preference, or whether both are independent consequences of a shared circuit architecture. ROI radius increases with depth ($r = +0.878$), raising the concern that larger ROIs at depth average over more neuropil and thus show artificially lower P ; however, the P -depth partial correlation controlling for radius strengthens slightly ($r = -0.757$ versus -0.706 at zero order), arguing against this confound. Calcium imaging with GCaMP6s introduces nonlinearities in the fluorescence-to-activity transformation that may differ across depth, though these nonlinearities would be expected to compress response ratios rather than to produce a systematic depth gradient. Layer boundaries were assigned based on depth from the pial surface without histological verification, though the depth range (140–518 μm) is consistent with the known extent of layer 2/3 in macaque V1 (Lund, 1988).

4.6 Future directions

The tissue from which these recordings were obtained is being prepared for serial electron microscopy reconstruction (Chatterjee et al., 2021). This will make it possible to compare the physiological properties of thousands of neurons—including the cross-orientation metrics reported here—with their reconstructed synaptic connectivity and to test directly whether the residual depth gradient in P that survives all measured confounds is accounted for by changes in inhibitory wiring architecture across layer 2/3.

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